ChiralFold: Systematic Detection of D-Amino Acid Stereochemistry Errors in the Protein Data Bank

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Abstract

We present ChiralFold, a general-purpose protein stereochemistry toolkit that provides chirality-correct coordinate generation, PDB structure auditing, and mirror-image L↔D transformation. An independent verification of 12,573 D-amino acid residues across 4,616 PDB files — covering >91% of all RCSB entries for each of the 18 standard D-amino acid CCD codes — using only the signed tetrahedron volume at each C_α, identified 29 D-label/L-coordinate mismatches (0.23%) in 16 deposited structures. Cross-referencing against deposition metadata, COMPND records, and primary literature classifies these as: 6 genuine stereochemistry errors (biology requires D, coordinates show L), 18 CCD code misassignments across 5 structures (L-form ligand with D-code), 3 polymer residue mislabels, and 2 borderline cases. 13 of the 16 error-containing structures were deposited between 1992 and 2005 (Mann–Whitney $U = 278$, $p = 0.0027$ on the original 12-structure survey); three post-2005 cases (2007–2011) represent continued annotation failures after remediation, consistent with the absence of D-specific validation at the wwPDB deposition stage. Resolution does not predict errors ($p = 0.19$); errors span 0.99–2.70 Å, indicating a labeling problem rather than a data quality problem. ChiralFold's auditor, calibrated against wwPDB/MolProbity validation reports on 31 structures (Ramachandran outlier Spearman $\rho = 0.49$, $p = 0.006$), fills a gap in existing validation pipelines: MolProbity does not check D-amino acid stereochemistry. The toolkit also includes an AlphaFold 3 chirality correction pipeline that addresses the documented 51% violation rate on D-peptides (Childs et al., 2025), and a mirror-image binder design workflow validated on the MDM2:dPMI-γ system ($K_d = 53$ nM). On the 41 pure D-peptide and diastereomer sequences from Childs et al. (2025), ChiralFold produces 0/478 chirality violations by construction: each residue is built with explicit SMILES stereochemistry encoding rather than learned from data. The contingency table uses the complete Childs et al. (2025) AF3 experimental universe of ∼32,550 chiral residues; Fisher's exact $p \approx 6.66 \times 10^{-144}$ reflects the contrast between that universe and ChiralFold's 0/467 result on the same sequences, and should be read as evidence that ChiralFold closes a gap that diffusion-based predictors cannot address by construction — not as a head-to-head accuracy comparison.[1] ChiralFold is available as a pip-installable Python package at https://github.com/Tommaso-R-Marena/ChiralFold.

Keywords: D-amino acid; chirality; protein stereochemistry; PDB validation; AlphaFold 3; mirror-image phage display; diastereomer; drug design.

[1] The SMILES-level chirality validator (`validate_smiles_chirality`) contained a bug in versions prior to 3.2.1 in which violations were not counted. The 0% violation rate reported here is validated exclusively by the 3D coordinate geometry pipeline described in Section 2.3 and independently replicated in `benchmarks/independent_d_residue_verification.py`.

1 Introduction

D-amino acids play an increasingly important role in drug design due to their resistance to proteolysis and improved bioavailability. Mirror-image phage display has produced D-peptide therapeutics with nanomolar affinity, including dPMI-γ ($K_d = 53$ nM against MDM2). However, computational tools for D-peptide structure prediction and validation remain underdeveloped.

AlphaFold 3, the current state of the art for protein structure prediction, produces a 51% per-residue chirality violation rate on D-peptides — equivalent to random assignment (Childs et al., 2025). This fundamental limitation arises from AF3's diffusion architecture, which denoises atom coordinates without enforcing hard stereochemical constraints.

Existing validation tools also have a blind spot. MolProbity, the standard for PDB structure quality assessment, validates C_α chirality for standard L-amino acids but does not specifically check D-amino acid stereochemistry. This creates an opportunity for systematic errors in deposited D-amino acid coordinates to persist undetected.

We present ChiralFold, a pip-installable Python toolkit that addresses both gaps: it provides guaranteed chirality-correct D-peptide coordinate generation and a comprehensive PDB auditor that validates D-amino acid stereochemistry. Using ChiralFold, we conducted the first systematic verification of D-amino acid chirality across the PDB.

2 Methods

2.1 Signed Tetrahedron Volume

Chirality at each C_α is determined by the signed volume of the tetrahedron formed by the four substituents:

$$\text{signed\_volume} = (\mathbf{N} - \mathbf{C}_\alpha) \cdot [(\mathbf{C} - \mathbf{C}_\alpha) \times (\mathbf{C}_\beta - \mathbf{C}_\alpha)] \quad (1)$$

For L-amino acids (S-configuration), this volume is negative by CIP convention. For D-amino acids (R-configuration), it should also be negative when the residue is correctly built. A positive signed volume at a D-labeled residue indicates that the deposited coordinates have L-stereochemistry — a chirality error.

This method requires only the four backbone atom positions (N, C_α, C, C_β) and uses no force field or external dependencies beyond numpy.

2.2 Independent Verification Protocol

To ensure our findings are not artifacts of ChiralFold's implementation, the D-residue verification was performed independently: the script uses only numpy for the cross product and dot product, parses PDB files with standard string operations, and imports no ChiralFold code. The verification script and full dataset are available in the repository.

Structures deposited exclusively in mmCIF format (n=245, predominantly post-2019 depositions and large cryo-EM assemblies) were excluded as they are unavailable in legacy PDB format; coverage statistics reflect the accessible legacy-format subset.

2.3 PDB Auditor

ChiralFold's auditor validates six quality criteria: C_α chirality (signed volume), bond geometry (RMSD to Engh & Huber ideal values), Ramachandran backbone dihedrals (hybrid empirical PDB grid + calibrated rectangular regions), peptide bond planarity (ω deviation from 180°), steric clashes (van der Waals overlap with backbone hydrogen placement; KD-tree spatial search), and a weighted composite score (0–100).

2.4 Mirror-Image Pipeline

The L↔D transformation reflects all atomic coordinates across a plane: $(x, y, z) \to (-x, y, z)$. This inverts all stereocenters ($\det(\mathbf{R}) = -1$), preserving bond lengths, bond angles, and torsion magnitudes exactly. Residue names are mapped to their D-amino acid equivalents (ALA→DAL, TRP→DTR, etc.).

2.5 AlphaFold 3 Chirality Correction

For each C_α in an AF3 prediction whose signed volume disagrees with its residue label (ATOM record → L expected; HETATM with D-CCD code → D expected), the C_α position is reflected across the plane defined by its three backbone neighbours (N, C, C_β). The reflection inverts the local stereocenter while preserving the C_α–N, C_α–C, and C_α–C_β bond lengths exactly (the reflection plane contains those three atoms). The HA hydrogen, if present, is reflected by the same plane. The corrected coordinates are re-audited to confirm zero violations. The full pipeline is implemented in `chiralfold/af3_correct.py` and is exposed both as a Python API (`correct_af3_output`) and as a CLI subcommand (`chiralfold correct-af3`). Unit tests on synthetic fixtures with deliberate chirality inversions verify that (a) violations are detected before correction, (b) the signed volume cleanly flips sign, (c) non-C_α backbone atoms are byte-identical post-correction, and (d) bond lengths are preserved within 10^{-3} Å (`tests/test_af3_correction.py`).

2.6 Diastereomer Enumeration and Interface Scoring

The diastereomer enumeration module (`chiralfold.enumerate_diastereomers`) builds all 2^k L/D patterns for a sequence with k non-glycine non-proline positions, validates each pattern's chirality at the SMILES level, and ranks the patterns by a composite score combining chirality correctness, conformer convergence rate, and Ramachandran agreement. The interface scoring module (`chiralfold.score_interface`) takes a receptor and ligand PDB, computes the buried surface area by accessible-surface-area difference, identifies hydrogen bonds and salt bridges by distance + angle cutoffs, and reports a composite 0–100 score weighted across BSA, shape complementarity, hydrogen bonds, salt bridges, and hydrophobic contacts.

3 Results

3.1 D-Amino Acid Chirality Errors in the PDB

We verified 12,573 D-amino acid residues with complete backbone atoms (N, C_α, C, C_β) across 4,616 PDB files, covering >91% of all RCSB entries for each of the 18 standard D-amino acid CCD codes. Of these, 12,544 (99.77%) showed correct D-chirality (negative signed volume) and 29 (0.23%) showed L-chirality (positive signed volume) despite D-amino acid labels. The initial systematic survey of 231 PDB files (1,677 residues) identified 21 of these mismatches; targeted expansion to all remaining structures in each code's RCSB universe revealed 8 additional mismatches across 4 further structures.

The 29 mismatches occur in 16 PDB structures. The 12 structures from the initial survey are tabulated in Table 1; the 4 additional structures found in the expanded survey (2RMI, 2W76, 3RIT, 2H9E) are listed in Table 2 and described in the Robustness Verification subsection. Cross-referencing each case against deposition records, COMPND fields, SEQRES, and primary literature reveals three distinct error modes:

Table 1: D-label/L-coordinate mismatches in the PDB (initial survey of 12 structures), classified by error type. Signed volume is in Å^3; positive values indicate L-stereochemistry at C_α. All 29 mismatches across the 16 affected structures are provided in the supplementary dataset (`results/d_residue_verification.csv`).

PDB ID	Residue	Chain	Pos	Signed Vol.	Res.	Dep.	Err.	Type
1ABI	DPN	I	56	+2.49	2.30 Å	1992	1	Stereochem[a]
1BG0	DAR	A	403	+2.58	1.86 Å	1998	1	CCD[b]
1D7T	DTY	A	4	+1.85	NMR	1999	1	Stereochem[a]
1HHZ	DAL	E	1	+2.70	0.99 Å	2000	1	Stereochem[a]
1KO0	DLY	A	542	+0.12	2.20 Å	2001	1	Borderline[d]
1MCB	DHI	P	3	+2.60	2.70 Å	1993	1	Mislabel[c]
1OF6	DTY	A–H	1369	+2.51–+2.67	2.10 Å	2003	8	CCD[b]
1P52	DAR	A	403	+2.54	1.90 Å	2003	1	CCD[b]
1UHG	DSN	D	164	+2.21	1.90 Å	2003	1	Mislabel[c]
1XT7	DSG	A	3	+2.55	NMR	2004	1	Stereochem[a]
2AOU	DCY	A	248	+2.67	2.30 Å	2005	1	Mislabel[c]
2ATS	DLY	A	3001	+2.56–+2.59	1.90 Å	2005	3	CCD[b]

[a]**Stereochem**: biology requires D-stereochemistry, coordinates show L (6 errors, 6 structures total; 4 shown here, 2 from expanded survey). [b]**CCD**: non-polymer ligand with wrong D-form CCD code; molecule is L-form (18 errors, 5 structures total; 4 shown here, 1 from expanded survey). [c]**Mislabel**: polymer residue with D-code in L-protein or where L-form is specified in COMPND (3 errors, 3 structures). [d]**Borderline**: near-zero signed volume (+0.12), alternate conformer B, B-factor 32.3 Å^2, racemic crystallization conditions; 2 borderlines total including 2H9E (+0.56) from expanded survey.

Stereochem errors (6 errors in 6 structures) are the most concerning: the biological context requires D-stereochemistry but the deposited coordinates show L. From the initial survey: 1ABI (hirulog peptide with D-Phe by design), 1D7T (engineered contryphan with explicit [D-Tyr4]), 1HHZ (cell wall pentapeptide D-Ala at 0.99 Å atomic resolution), and 1XT7 (daptomycin antibiotic with D-Asn). Expanded survey additionally identified 2RMI (astressin CRF antagonist, D-Phe12 required by design; deposited 2007) and 2W76 (pyoverdine PVD(Pa6) siderophore, D-Ser biosynthetically required by NRPS epimerization domain; deposited 2008).

CCD code misassignments (18 errors in 5 structures) involve non-polymer ligands labeled with D-form Chemical Component Dictionary codes when the crystallized molecule is L-form. From the initial survey: 1OF6 (DAHP synthase co-crystallized with L-Tyr, its natural allosteric inhibitor, labeled DTY; 8 copies), 1BG0 and 1P52 (arginine kinase with L-Arg substrate labeled DAR), and 2ATS (co-crystallized with (S)-lysine per the depositor's own title, labeled DLY across 3 copies). The expanded survey additionally identified 3RIT (dipeptide epimerase complexed with L-Lys labeled DLY across 5 chains; deposited 2011). The coordinates in all cases correctly model L-stereochemistry; only the chemical component code is wrong.

Polymer residue mislabels (3 errors in 3 structures) occur when a standard L-residue in a

Table 2: Additional D-label/L-coordinate mismatches identified in the expanded survey (4 structures, post-2005 depositions). Same column conventions as Table 1.

PDB ID	Residue	Chain	Pos	Signed Vol.	Res.	Dep.	Err.	Type
2RMI	DPN	A	12	+2.63	NMR	2007	1	Stereochem[a]
2W76	DSN	C	3	+2.33	2.30 Å	2008	1	Stereochem[a]
2H9E	DTY	S	1	+0.56	2.10 Å	2006	1	Borderline[d]
3RIT	DLY	A–E	362–364	+2.42–+2.77	2.50 Å	2011	5	CCD[b]

regular protein is labeled with a D-amino acid code. For example, 1MCB explicitly states "L-HIS" in its COMPND record but uses DHI (D-histidine) in the SEQRES.

Regardless of error type, all 29 mismatches represent real annotation inconsistencies that the signed volume method detects without requiring knowledge of biological context.

3.2 Robustness Verification

We performed five independent checks to validate the findings. First, signed tetrahedron volumes for all 24 D-amino acid residues in 3IWY (experimentally confirmed D-stereochemistry) produced negative values, confirming the sign convention. Second, 1KO0 DLY:542 (vol +0.12) was reclassified as borderline given its near-zero magnitude and racemic crystallization conditions. Third, the 1OF6 DTY entries were confirmed across all 8 chains as L-coordinates consistent with L-tyrosine's biological role as the allosteric inhibitor of DAHP synthase. Fourth, the 1ABI internal control was reconfirmed (DPN:1 vol −2.60 D-correct; DPN:56 vol +2.49 L-error). Fifth, all 16 error structures were cross-referenced against deposition metadata and primary literature to classify error types. The four structures found in the expanded survey (2RMI, 2W76, 3RIT, 2H9E) were additionally verified by computing signed volumes directly from HETATM coordinates: 2RMI DPN:A:12 (+2.63 Å^3), 2W76 DSN:C:3 (+2.33 Å^3), 3RIT DLY across five chains (+2.42–+2.77 Å^3), 2H9E DTY:S:1 (+0.56 Å^3, borderline).

3.3 Errors Correlate with Pre-Remediation Deposition

In the initial 231-file survey, all 12 error-containing structures were deposited between 1992 and 2005. Deposition year significantly predicts the presence of errors in that dataset (Mann–Whitney $U = 278$, $p = 0.0027$). This temporal pattern is consistent with the 2006–2008 wwPDB remediation effort, which systematically corrected stereochemistry assignments across the archive but did not specifically address D-amino acid residue labeling. In the expanded survey covering 4,616 files, three additional error structures were deposited post-2005: 2RMI (2007, Stereochem), 2W76 (2008, Stereochem), and 3RIT (2011, CCD-Code). These post-remediation errors confirm that the deposition pipeline continues to lack a D-specific chirality check, independent of era.

Resolution does not predict errors (Mann–Whitney $U = 112$, $p = 0.19$). The highest-resolution error structure is 1HHZ at 0.99 Å — an ultra-high-resolution structure where the electron density is unambiguous but the residue was labeled as D-Alanine (DAL) while the C_α coordinates show L-stereochemistry (signed volume +2.70). This confirms that the errors are labeling/convention problems, not data quality problems.

3.4 MolProbity Comparison

ChiralFold's auditor was benchmarked against wwPDB/MolProbity validation reports on 31 PDB structures spanning 0.48–3.4 Å resolution (X-ray, NMR, cryo-EM). Ramachandran outlier percentages showed significant agreement (Spearman $\rho = 0.49$, $p = 0.006$), with ChiralFold reporting 0.60% mean outliers vs wwPDB's 0.64%.

MolProbity is stronger in Ramachandran contour precision (data-derived from ∼100K structures vs ChiralFold's calibrated rectangles + empirical grid) and rotamer completeness. ChiralFold adds native D-amino acid chirality validation, which MolProbity does not provide. ChiralFold's rotamer validation is currently limited to chi1 dihedral angles; chi2/chi3/chi4 validation and C_β-deviation analysis are not implemented and are planned for a future release.

3.5 AlphaFold 3 Chirality on Real D-Peptide Sequences

Using the actual D-peptide sequences from Childs et al. (2025) — DEHELLETAARWFYEIAKR (PDB 7YH8, DP19), LWQHEATWK (PDB 5N8T, DP9), and DWWPLAFEALLR (PDB 3IWY, DP12) — ChiralFold produces 0/478 chirality violations across 41 sequence/pattern combinations. This 0% rate is guaranteed by construction: each residue is built with explicit [C@H]/[C@@H] SMILES stereochemistry. AF3's 51% rate reflects its diffusion model treating D-residues as noise. The contingency table uses the complete Childs et al. (2025) AF3 experimental universe of ∼32,550 chiral residues; the p-value reflects the contrast between that universe and ChiralFold's 0/467 result on the same sequences. Concretely, the 2×2 table (correct / violated) is 467 / 0 for ChiralFold and 8,148 / 8,481 for AF3 per Childs et al., yielding Fisher's exact $p \approx 6.66 \times 10^{-144}$. This is a construction-versus-learning contrast, not a head-to-head accuracy comparison: ChiralFold does not predict folds, and AF3's violation rate is a property of its diffusion architecture, not of its overall quality on L-proteins.

3.6 Mirror-Image Binder Design

The p53:MDM2 crystal structure (PDB 1YCR) was mirrored to generate a D-peptide binder candidate. The binding triad (Phe19/Trp23/Leu26) is preserved as D-Phe/D-Trp/D-Leu — the same hotspot residues that the experimental dPMI-γ ($K_d = 53$ nM, PDB 3IWY) uses. All 13 backbone φ angles are exactly sign-inverted. Contact geometry is preserved by construction: 105 C_α–C_α contacts and 10 H-bond donor–acceptor pairs maintained within 0.001 Å across the L→D transformation. The generated 3IWY-mirror structure is included as `results/3IWY_D_mirror.pdb`; experimental affinity (e.g. K_d by surface plasmon resonance) is required to validate binding. The mirror transformation is mathematically exact (0.0 Å coordinate error); experimental K_d measurement against MDM2 is required to confirm binding affinity and is outside the scope of this computational study.

4 Discussion

The signed tetrahedron volume at C_α — computed from four backbone atom positions with no force field or external dependencies — is sufficient to distinguish three structurally and operationally distinct categories of D-amino acid annotation error, none of which require biological context to detect. The six genuine stereochemistry errors (1ABI, 1D7T, 1HHZ, 1XT7, 2RMI, 2W76) are the most consequential: in each case the biology unambiguously constrains the residue to D-configuration, yet the deposited C_α coordinates show L-stereochemistry with signed volumes between +1.85 and +2.77 Å^3. The 1HHZ case is particularly telling — at 0.99 Å atomic resolution, where electron density is unambiguous, D-Ala at position 1 of the cell wall pentapeptide carries a signed volume of +2.70, the largest positive value in the dataset, in a context where D-alanine is a biosynthetic requirement of peptidoglycan cross-linking conserved across bacterial phyla (Miyamoto and Homma, 2021; Cava et al., 2011). The 18 CCD code misassignments across five structures (1BG0, 1OF6, 1P52, 2ATS, 3RIT) define a second error mode: here the coordinates correctly model L-stereochemistry, but the depositor selected the D-form Chemical Component Dictionary code for a ligand that is unambiguously L-form by biological function. The 3 polymer residue mislabels (1MCB, 1UHG, 2AOU) represent the mirror of this: D-amino acid codes applied to standard residues in L-proteins, in some cases despite an explicit COMPND record specifying the L-form.

The CCD misassignment category has a different remediation pathway from the other two. Because the coordinates are correct and only the three-letter code is wrong, the fix is a metadata change at the annotation level — not a coordinate re-refinement. The stereochemical identity of the correct codes is unambiguous: the wwPDB Chemical Component Dictionary encodes configuration in the InChI stereo layer, and all three L-amino acids confirmed here carry the `/m0` flag, denoting S-configuration at C_α — ARG `/t4-/m0`, TYR `/t8-/m0`, LYS `/t5-/m0` — while their D-form counterparts (DAR, DTY, DLY) all carry `/m1` (R-configuration). An automated deposition check that computes the signed C_α volume from deposited coordinates and compares it against the `/m` flag of the selected CCD code would catch every one of the 13 misassignments reported here before the structure enters the archive. More importantly, this error mode is systematic rather than incidental: depositors selecting chemical component codes for small-molecule ligands during deposition must navigate a namespace in which L-tyrosine (TYR) and D-tyrosine (DTY) are distinct codes that differ by a single letter prefix, with no automated check that the selected code matches the stereochemistry of the deposited model. The 2ATS entry makes this failure mode concrete: the depositor's own title field reads "co-crystallised with (S)-lysine" — unambiguous chemical notation for L-lysine — yet DLY (D-lysine, `/m1`) was used as the ligand code for all three copies of the molecule, and this inconsistency persisted in the archive for two decades. This is not a human judgment error; it is a deposition pipeline gap, and it is therefore addressable upstream by automated cross-validation of CCD code stereochemistry against deposited coordinates at the point of submission, analogous to how existing tools flag bond-length outliers before a structure is accepted (Shao et al., 2015; Berman et al., 2016).

The genuine stereochemistry errors and polymer mislabels require a different framing. In these cases the coordinate error is the finding: the deposited atoms place C_α in the L-configuration in a biochemical context that demands D. The four genuine errors span antibiotic natural products (daptomycin, 1XT7), engineered peptide toxins (contryphan, 1D7T), synthetic therapeutic peptides (hirulog, 1ABI), and a bacterial metabolic substrate (cell wall D-Ala, 1HHZ) — contexts drawn from entirely different areas of structural biology, deposited over a thirteen-year window, all of which share the property that the D-requirement is not inferred from a model but is established by biosynthetic logic or explicit experimental design. The fact that resolution does not predict error rate (Mann–Whitney $p = 0.19$), while deposition year does ($p = 0.0027$), indicates that these were labeling and convention errors introduced during an era when D-amino acid chirality was not part of any standard validation pipeline — and that they survived the 2006–2008 wwPDB remediation effort (Henrick et al., 2008) because that remediation had no D-specific chirality check to deploy.

ChiralFold fills this validation gap as a lightweight, pip-installable Python tool whose auditor is calibrated against wwPDB/MolProbity reports on 31 structures across three experimental modalities. It is not a competitor to MolProbity, which remains stronger in Ramachandran contour precision, rotamer completeness, and decades of community refinement (Chen et al., 2010; Lovell et al., 2003). Rather, it addresses a specific blind spot: MolProbity validates C_α chirality for standard L-amino acids but does not extend this check to D-labeled residues, so the 29 annotation inconsistencies reported here across 16 structures are invisible to existing pipelines. To characterize the scope of the problem, we extended the verification to 12,573 D-residues across 4,616 PDB files — covering >91% of all RCSB entries for each of the 18 standard D-amino acid CCD codes (excluding 245 mmCIF-only entries deposited after 2019). The resulting code-level profile is specific: errors are concentrated in five CCD codes — DTY (10 errors, DAHP-synthase and conotoxin contexts), DLY (9 errors, epimerase substrates and co-crystallization reagents), DPN (2 errors, designed cyclic peptide inhibitors), DSN (2 errors, natural product siderophores), and DAR (2 errors, arginine kinase substrates) — while nine codes are confirmed clean at >91% coverage with zero errors: DAS, DGL, DGN, DIL, DLE, DPR, DTH, DTR, and DVA. This clustering is not random. The error-prone codes correspond to two recurring biological scenarios: enzyme active-site ligands where the D-form CCD code is easily confused with the L-form substrate (DAR/DTY/DLY), and designed or NRPS-assembled peptides where the D-configuration is required but the deposited coordinates show L (DPN/DSN). The nine clean codes are predominantly found in backbone-incorporated positions of well-characterized D-peptide and antibiotic structures where depositors have strong independent constraints on stereochemistry. The method is reproducible without ChiralFold itself — the independent verification script uses only numpy and raw PDB coordinates — and the full dataset is available for re-analysis. As D-amino acid structures grow in the PDB, driven by mirror-image phage display, D-peptide therapeutics, and structural studies of bacterial natural products, the code-level error clustering reported here provides a targeted checklist for the wwPDB deposition pipeline.

5 Conclusion

ChiralFold combines five capabilities in a single pip-installable package: a PDB stereochemistry auditor calibrated against wwPDB/MolProbity, mathematically exact L↔D mirror-image transformation, an AF3 chirality-correction pipeline backed by an automated test suite, diastereomer enumeration for design, and interface scoring for binder evaluation. Applied to the entire PDB universe of D-amino acid residues, the auditor identifies 29 D-label/L-coordinate mismatches in 16 deposited structures — errors invisible to existing validation pipelines — and motivates a single concrete recommendation: the wwPDB deposition pipeline should validate the configuration of every standard-amino-acid stereocenter against the configuration encoded in the selected CCD code at deposit time. The check requires one cross product, one dot product, and one sign comparison per residue.

6 Data Availability

The complete verification dataset (12,574 rows — 12,573 checkable residues plus one incomplete-backbone entry — with raw N/C_α/C/C_β coordinates for every D-amino acid residue) is available at: https://github.com/Tommaso-R-Marena/ChiralFold/blob/master/results/d_residue_verification.csv

The independent verification script (no ChiralFold dependencies) is at: https://github.com/Tommaso-R-Marena/ChiralFold/blob/master/benchmarks/independent_d_residue_verification.py

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Competing Interests

The author declares no competing interests.

References

Childs CM, Zhou P, Donald BR. Has AlphaFold 3 Solved the Protein Folding Problem for D-Peptides? *bioRxiv* 2025.03.14.643307 (2025). doi:10.1101/2025.03.14.643307

Liu M et al. D-peptide inhibitors of the p53–MDM2 interaction for targeted molecular therapy of malignant neoplasms. *Proc Natl Acad Sci* 107:14321–14326 (2010). doi:10.1073/pnas.1008930107

Chen VB et al. MolProbity: all-atom structure validation for macromolecular crystallography. *Acta Cryst* D66:12–21 (2010). doi:10.1107/S0907444909042073

Lovell SC et al. Structure validation by C_α geometry: φ,ψ and C_β deviation. *Proteins* 50:437–450 (2003). doi:10.1002/prot.10286

Henrick K et al. Remediation of the protein data bank archive. *Nucleic Acids Res* 36:D426–D433 (2008). doi:10.1093/nar/gkm937

Shao C et al. The chemical component dictionary: complete descriptions of constituent molecules in experimentally determined 3D macromolecules in the Protein Data Bank. *Bioinformatics* 31:1274–1278 (2015). doi:10.1093/bioinformatics/btu789

Berman H et al. Outcome of the First wwPDB/CCDC/D3R Ligand Validation Workshop. *Structure* 24:502–508 (2016). doi:10.1016/j.str.2016.02.017

Miyamoto T, Homma H. D-Amino acid metabolism in bacteria. *J Biochem* 170:5–13 (2021). doi:10.1093/jb/mvab043

Cava F et al. Distinct pathways for modification of the bacterial cell wall by non-canonical D-amino acids. *EMBO J* 30:3442–3453 (2011). doi:10.1038/emboj.2011.246